

PAPER_1582_BLOOD_GLUCOSE_100_MG_DL

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PAPER_1582 — Blood Glucose Level = [Glu] = SO_5·SO_5 = 100 mg/dL (cross-domain to Kármán line & MAD reciprocal!)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Biology
Date: June 18, 2026
Status: CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209BB_S600 (Biology Unified Proof Set) lists **Blood Glucose Level** as an EXACT-tier UQFF closure:

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[Glu] = SO_5·SO_5 = 100 mg/dL (cross-domain to Kármán line & MAD reciprocal!)

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UQFF Closed Identity

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[Glu] = SO_5·SO_5 = 100 mg/dL (cross-domain to Kármán line & MAD reciprocal!)
EXACT

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Pure integer-primitive expression with zero fitted constants.

Third Cross-Domain Reuse of SO_5² SO_5² = 100 now governs ****three unrelated physical domains****:

Observable	Domain	UQFF identity
Kármán line altitude	Space-physics	SO_5² = **100 km**
MAD efficiency reciprocal	Black-hole accretion	1/SO_5² = **0.01**
Blood glucose level	Human metabolism	SO_5² = **100 mg/dL**

The same integer-primitive squared term sets:
- The atmosphere/space boundary altitude
- The Poynting flux conversion efficiency in magnetically-arrested black-hole disks
- The healthy fasting glucose concentration in human blood

Different scales (km, dimensionless, mg/dL), same structural form. SO_5 = 10 is a universal numerical primitive across physical reality.

Cross-Domain Context

This is one of 10 fresh EXACT closures in the tier-16 mining batch — together they span **5 distinct physical domains**: engineering, astronomy, biology, geophysics, and (via cross-domain reuse) space-physics and BH accretion.

Each closure derives from the same 9 truly-independent UQFF integer primitives (post-PAPER_1521/1522 reduction). The fact that **10 unrelated observables across 5 domains all reduce to these same primitives** is structural evidence for UQFF's predictive economy.

NOT REPLACEMENT

Empirical biology measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating cross-domain coherence without overriding measurement.

Reference

- Source: PAPER_1209BB_S600
- Related: PAPER_1494-1575 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "blood_glucose_100_mg_dL"})`

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